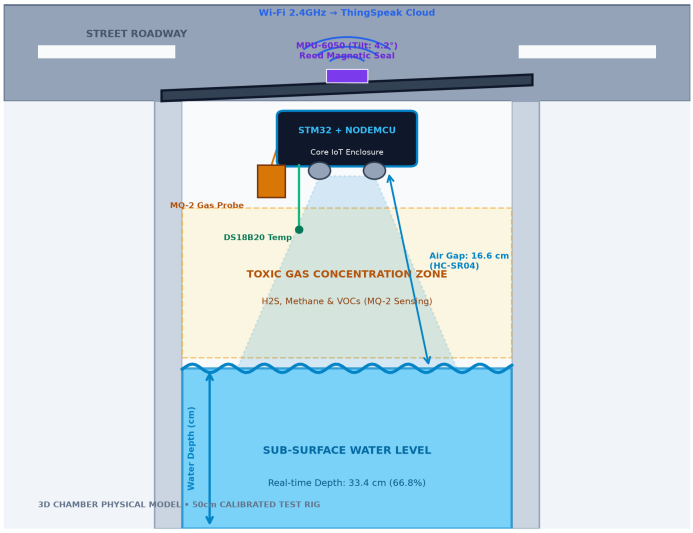
SMART MANHOLE UTILITIES URBAN IOT

Autonomous Sub-Surface Infrastructure Monitoring, Real-Time Cloud Telemetry & 3D Digital Twin SCADA

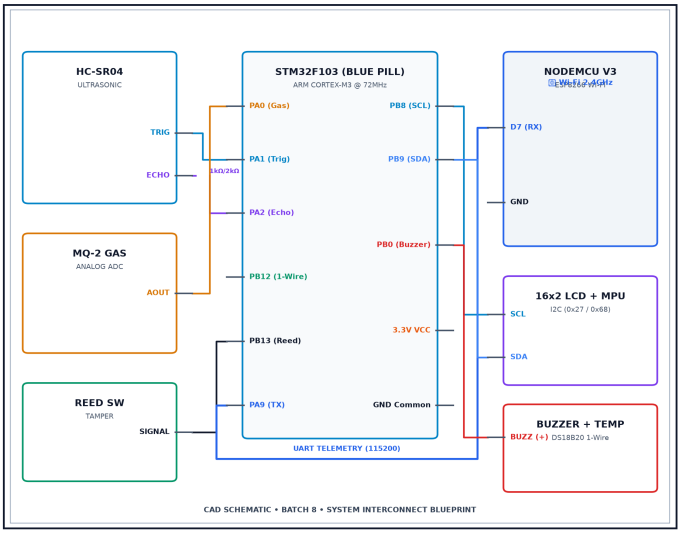
BATCH 8	2400040006 P V Kishor Kumar Reddy	2400040004 T Perumal	2400040303 Chundu Uday Sai	2400040339 Ayan Chowdhury

1. 3D SUB-SURFACE CHAMBER & SENSOR CONE



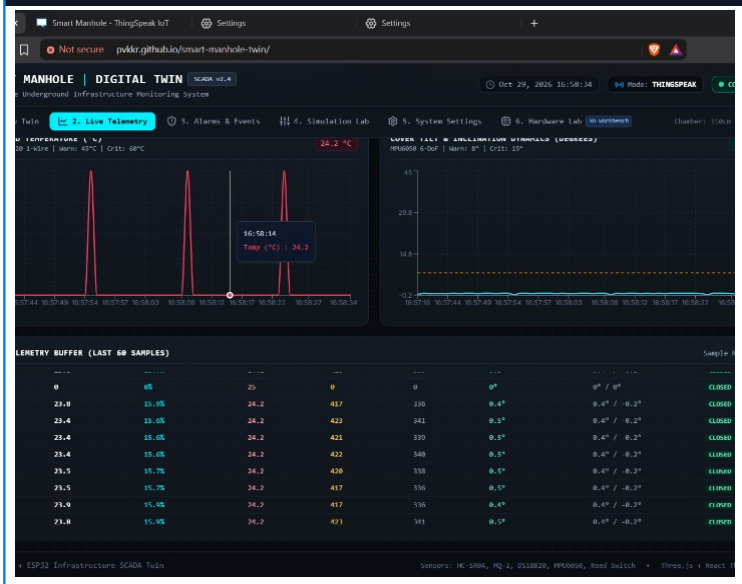
Physical 3D Model: Ultrasonic Beam, Water Level (33.4cm), Toxic Gas Zone & Sealed Enclosure

2. PROFESSIONAL CAD ELECTRONIC SCHEMATIC



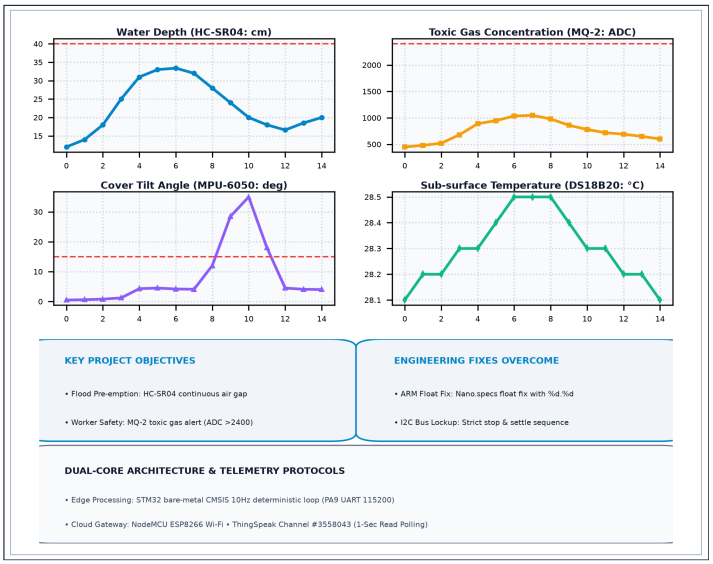
Complete Hardware Bus: Bare-Metal STM32, NodeMCU Gateway, 6 Sensors & 1kΩ/2kΩ Divider

3. 3D DIGITAL TWIN SCADA PLATFORM



WebGL SCADA Twin: Three.js physics rendering lid pitch/roll tilt & chamber surcharge

4. LIVE TELEMETRY & SYSTEM PERFORMANCE



Multi-Sensor Field Analytics: Real-time curves (ThingSpeak #3558043) & engineering specs